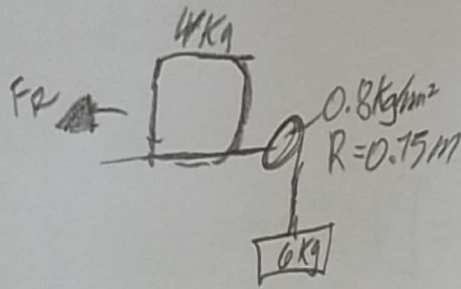


Primer Examen Parcial

1)



$$\sum F_y = 0$$

$$N = mg = 0$$

$$N = mg = (4)(9.8) = 39.2 \text{ N}$$

$$\text{Fricción} = F_r$$

$$F_r = \mu N_r = \frac{3}{10} (39.2 \text{ N}) = 11.76 \text{ N}$$

$$\sum F_x = ma \quad T_2 - 11.76 \text{ N} = 4a$$

Polea

$$\sum \tau = I \alpha$$

$$T_1 (0.25 \text{ m}) - T_2 (0.25 \text{ m}) = (0.8 \text{ kg/m}^2) \frac{a}{0.25 \text{ m}}$$

$$T_1 - T_2 = \frac{(0.8)a}{0.25} = (3.2 \text{ kg})a$$

6 kg

$$\sum F_y = ma$$

$$mg - T_1 = ma$$

$$(6 \text{ kg})(9.8 \text{ m/s}^2) - T_1 = (6 \text{ kg})a$$

3)
Primera masa

$$T_2 = (4) a + 11.76 \text{ N}$$

$$T_1 - T_2 = (12.8) a$$

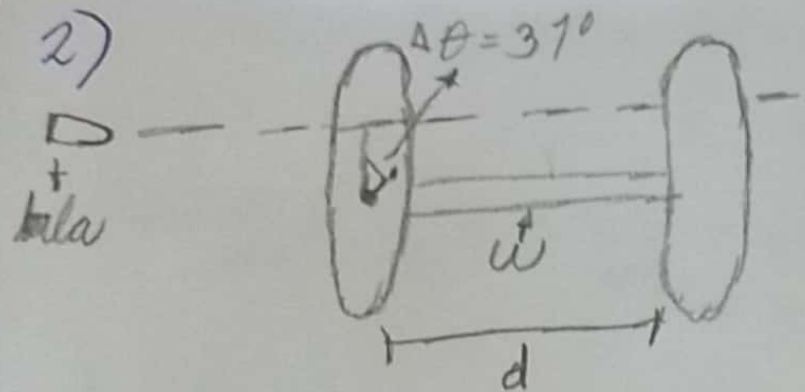
$$(58.8 \text{ N} - (6) a) - [(4) a + 11.76 \text{ N}] = (12.8 \text{ kg}) a$$

$$58.8 \text{ N} - 6a - 4a - 11.76 \text{ N} = (12.8 \text{ kg}) a$$

$$58.8 \text{ N} - 11.76 \text{ N} = 12.8 a + 6a + 4a$$

$$47.04 \text{ N} = 22.8 a$$

$$a = 2.06 \text{ m/s}^2$$



$$\omega = \frac{\Delta\theta}{t}$$

$$V = \frac{d}{t}$$

$$t = \frac{\Delta\theta}{\omega}$$

$$V = \frac{\omega d}{\Delta\theta}$$

$$d = 80 \text{ cm}$$

$$V = \frac{900 \frac{\text{rev}}{\text{min}} \cdot \frac{360^\circ}{1 \text{ rev}} \cdot \frac{1 \text{ min}}{60 \text{ s}} \cdot 0.8 \text{ m}}{37^\circ}$$

$$80 \text{ cm} \cdot \frac{1 \text{ m}}{100 \text{ cm}} = 0.8 \text{ m}$$

$$V = 139.35 \text{ m/s}$$

3)

Disco Grande = $M = \text{masa}$
 $R = \text{radio}$

Disco pequeño = $m = \frac{2}{3} M$

$$r = \frac{2}{5} R$$

$$250 \frac{\text{rev}}{\text{min}} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}} \cdot \frac{1 \text{ min}}{60 \text{ s}} = 26.18 \text{ rad/s}$$

$$L = I\omega$$

$$I_1\omega_1 + I_2\omega_2 = I_2\omega_{f2} + I_2\omega_{f2}$$

$$I_1\omega_1 = \omega_f (I_1 + I_2)$$

$$\frac{1}{2} MR^2 (26.18) = \omega_f \left(\frac{1}{2} MR^2 + \frac{4}{75} MR^2 \right)$$

$$13.09 = \omega_f (0.553)$$

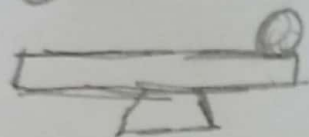
$$\omega_f = \underline{23.67 \text{ rad/s}}$$

$$I_1 = \frac{1}{2} MR^2$$

$$I_2 = \frac{1}{2} \left(\frac{2}{3} M \right) \left(\frac{2}{5} R \right)^2$$

$$I_2 = \frac{4}{75} MR^2$$

4)



$$V = \sqrt{2gh} = \sqrt{2(9.8)(12)} = 15.34 \text{ m/s}$$

M. angular

$$(5 \text{ kg})(15.34 \text{ m/s})(2 \text{ m}) = (5 \text{ kg})V_A(2 \text{ m}) + I\omega + (5 \text{ kg})V_B(2 \text{ m})$$

$$I = \frac{1}{12}(8)(4)^2 = 10.667 \text{ kg/m}^2$$

$$V_A = V_B = \omega r = V$$

$$\omega = \frac{V}{2 \text{ m}}$$

$$(5)(15.34)(2) = (5)(2)V + (10.667)\frac{V}{2} + (5)(2)V = 153.4 \text{ kg/m}^2 \text{ s} =$$

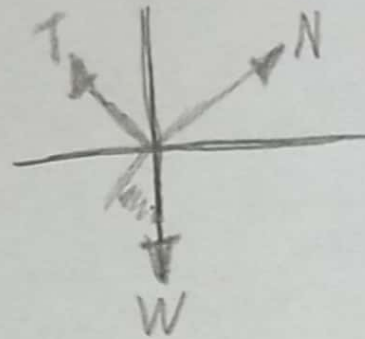
$$(25.333 \text{ kg/m})V \quad V = 6.055 \text{ m/s}$$

$$h = \frac{V^2}{2g} = \left(\frac{6.055^2}{2(9.8)} \right) = \underline{1.87 \text{ m}}$$

$$5) 12,000g \div \frac{1kg}{1000g} = 12kg$$

$$R = 0.1m$$

$$100mm \div \frac{1cm}{10mm} = 10cm$$



$$\sum F_x = \sin 37^\circ W - T = ma$$

$$I \alpha = Fx$$

$$I \left(\frac{a}{R} \right) = TR$$

$$I \left(\frac{2}{0.1} \right) = 46.77(0.1)$$

$$20I = 4.677$$

$$I = 0.23 kg m^2$$

$$\begin{aligned} \sin 37^\circ (12)(9.8) - T &= 12(2) \\ 70.77 - T &= 24 \\ T &= 46.77 N \end{aligned}$$